

VBF-C1T1R1 datasheet - Virtual Battery Factory deliverable

Cell Datasheet - R4-V3-margin (champion)

VBF-C1T1R1-DSH-001 | Case c1_t1_r1 | Virtual Battery Factory

General

| Item | Value | Source |

|---|---|---|

| System | NMC811 / graphite (Chen2020 base), 4.2 V upper cut-off | parameter dump |

| Cell area | 0.1027 m² (0.065 x 1.58 m) | calc-energy output |

| Cell mass | 35.603 g | calc-energy output |

| Cell thickness | 182.8 um | calc-energy output |

| 1C current | 5 A | contract |

Electrical

| Item | Value |

|---|---|

| Contract capacity | 5.0306 Ah |

| Contract energy | 17.8975 Wh |

| Gravimetric energy density | 502.70 Wh/kg (threshold 392.61: PASS) |

| Volumetric energy density | 953.34 Wh/L |

| Midpoint voltage | 4.0073 V |

| DCR | 0.14679 mOhm |

| Power density | 797.3 kW/kg |

Fast charge (4C = 20 A, 45 C ambient, SPMe + lumped thermal)

| Item | Value |

|---|---|

| T_max | 323.98 K = 50.83 C (threshold 60 C: PASS) |

| Anode potential minimum | +0.0503 V (>= 0 V: no lithium plating) |

Abuse tolerance

| Item | Value |

|---|---|

| Overcharge protocol (0.5C to upper cut-off + 0.5 V) | reached 4.70 V |

| Overcharge T_max | 299.45 K |

| Thermal runaway model (run-tr --sim coupling, mass 0.0356025 kg) | triggered = false; T_max 299.45

Key design parameters (overrides over Chen2020)

Positive CC 8 um (Al); negative CC 6 um (Cu); separator 8 um / porosity 0.55;

t+ 0.7; D_e 2e-9 m²/s; kappa 3.0 S/m; negative particle radius 3 um; h = 100 W/m²/K.